Northwestern University

Final Project Report
"Classroom Engagement Detection"

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1 Project Goal

The goal of this project is to design a system that can estimate **student engagement** during online lectures. Since it is difficult for instructors to understand how attentive students are in remote settings, our approach focuses on recognizing visible behaviors related to engagement. Examples include **nodding**, **neutral expressions**, **frowning**, **yawning**, **using a phone**, **drinking**, **head tilting**, and **hand raises**. These behaviors were selected because they appear frequently in online classes and can be detected clearly through webcam video.

The system provides a prediction for each student and then combines these predictions into an overall engagement estimate. This is intended to help instructors adjust their pacing or explanations in real time. Because our dataset is small and recorded in a controlled environment, we focused on a **lightweight** and **CPU-friendly** method that runs reliably without high-end hardware.

2 Method and Technical Approach

Our approach follows a **seven-stage pipeline**: data collection, data preprocessing, feature extraction, sequence modeling, model training, evaluation, and deployment. Each stage contributes to the final real-time engagement system.

2.1 Data Collection

Data was collected over two weeks from ten participants within our team and from friends. The dataset consists of Zoom meeting recordings, which were chosen because they simulate a realistic online learning environment. All footage was manually labeled to identify activities linked to engagement levels. These actions include both positive indicators such as **neutral expressions**, **hand raises**, **nodding**, and **frowning**, and negative indicators such as **yawning**, **phone use**, **head tilting**, and **drinking**.

Two lighting conditions were recorded (normal and low light), adding variability to the dataset even though the difference was sometimes subtle. In total, the dataset contains approximately **32,000 samples** based on ten participants, two environments, eight activities, and roughly two hundred frames per activity. All recordings were performed voluntarily with consent, and participants followed a simple script to ensure clear and consistent labeling.

2.2 Data Preprocessing

Raw video recordings were processed to isolate the relevant information needed for analysis. Most actions were converted into individual JPG frames, while the nodding action was preserved as MP4 clips to avoid losing temporal motion. All frames were converted to **RGB color space** and resized to match the input requirements of MediaPipe.

MediaPipe occasionally failed to detect landmarks because of occluded faces or extreme head angles. When this occurred, these frames were removed instead of corrected, since inaccurate interpolation would harm training. To reduce redundancy, we also applied a simple **frame skipping** strategy in actions with minimal movement. This reduced dataset size, improved balance across behaviors, and lowered computational cost.

2.3 Feature Extraction

To convert raw video frames into usable numerical data, we used **MediaPipe Holistic**, which extracts 468 face landmarks and 33 pose landmarks. This tool was ideal for our project because it operates in real time on CPU hardware and does not require training a deep learning model.

From these raw landmarks, we computed several **geometric features** that are informative for engagement-related behaviors. These include:

- **Head Pose** (pitch, yaw, roll) estimated using a PnP algorithm.
- **Eye Aspect Ratio (EAR)** to estimate eyelid openness.
- **Mouth Aspect Ratio (MAR)** to capture yawning or speaking motions.
- **Eyebrow distance** for detecting frowning or raised eyebrows.
- **Wrist-shoulder height difference** to detect hand raises or drinking.

These engineered features summarize the most relevant aspects of engagement and are more stable than raw pixels, especially given our relatively small dataset.

2.4 Sequence Modeling

Many behaviors in our dataset involve changes over time, so we incorporated temporal context through two sequence modeling approaches.

The first version relies on a **statistical window** that stores a short sequence of recent frames and computes simple statistics such as means and standard deviations. We also applied an **exponential moving average (EMA)** to smooth noise. This approach is computationally inexpensive, robust with limited data, and well-suited for sustained behaviors like phone use or hand raises.

The second version uses an **LSTM** to process short sequences of feature vectors directly. This method captures motion patterns more accurately, especially for behaviors like nodding. However, with our dataset size, the LSTM was more affected by overfitting and required careful parameter tuning. Although both models were tested, the statistical version offered more stable real-time performance, while the LSTM provided a useful comparison for understanding the limits of our dataset.

2.5 Model Training and Evaluation

A **Random Forest classifier** was used to map sequence-aware features to one of the eight activity labels. This model was chosen because it performs well on geometric feature sets, handles small datasets reliably, and allows inspection of feature importance. The classifier receives roughly twenty engineered features from the sequence modeling step.

Training used an **80/20 train–test split**, with shuffled data to maintain class balance. Because the dataset is small, some participants appear in both sets, meaning evaluation mainly measures action-level generalization rather than subject-level generalization. A future extension could evaluate the system on unseen participants to measure robustness more strictly.

Model performance was assessed using **accuracy**, **precision**, **recall**, and **F1-score**. A confusion matrix helped identify common confusions, such as nodding versus subtle head tilting, likely due to similar landmark movement. These insights led to adjustments in smoothing and feature selection. Overall, evaluation confirmed that geometric features combined with lightweight models are suitable for recognizing the target activities.

2.6 Deployment

The complete system was deployed as a Python script using **OpenCV** for real-time video handling. For each frame, the system extracted MediaPipe landmarks, computed geometric features, updated the temporal buffer, and produced a prediction using the Random Forest classifier. In the statistical version, additional heuristics improved stability. For example, consistently high MAR values reinforced yawning predictions to avoid rapid fluctuations.

Predicted activities are displayed directly on the video feed, giving users immediate feedback. The system also computes a simple **engagement score** by mapping each action to a positive or negative weight and averaging across students. These weights can be adjusted based on instructor preferences. Performance decreases when faces rotate significantly or lighting is too dim, which are limitations of the current methodology.

3 YouTube Demo

A demonstration video showing the full system in operation is available on YouTube:

https://www.youtube.com/watch?v=fZIWC5KLpak